Linear regression, the normal distribution, and confidence intervals

How the three concepts fit together

You're noticing something very important: **linear regression, the normal distribution, and confidence intervals are three different mathematical ideas.** We combine them because each solves a different part of one problem.

Coming from a calculus and probability background, I think the clearest way to understand this is to derive the entire process, rather than memorize the regression formulas.

We'll answer three questions:

1. Why do we fit a line to data?
2. Where does the normal distribution suddenly come from?
3. How does the normal distribution allow us to construct a confidence interval for the slope?

1. First, forget probability: fit a straight line

Imagine you're studying how many points a student scores depending on how many hours they study.

You collect five observations:

Hours studied x	Exam score y
1	2
2	3
3	5
4	4
5	6

You suspect the relationship is approximately linear:

$$y \approx a + bx$$

Here, b represents how many additional points we expect for one additional study hour.

We want to find the best a and b.

Notice how the best-fitting line changes when observations change. It is chosen to minimize the squared vertical distances between the points and the line.

We do this using **ordinary least squares**:

$$\min_{a,b} \sum_{i=1}^{n} (y_i - a - bx_i)^2$$

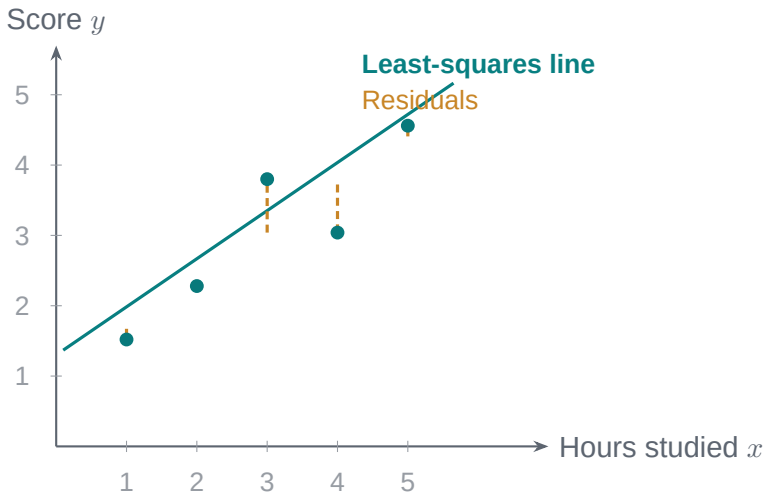


Figure 1: A static view of the least-squares idea: the fitted line and the vertical residuals. When observations change, the best-fitting line generally changes too.

By taking partial derivatives, we obtain:

$$\hat{b} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$\hat{a} = \bar{y} - \hat{b}\bar{x}$$

For our five observations:

$$\bar{x} = 3, \qquad \bar{y} = 4$$

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = 9$$

$$\sum(x_i - \bar{x})^2 = 10$$

Therefore:

$$\hat{b} = \frac{9}{10} = 0.9$$

$$\hat{a} = 4 - (0.9)(3) = 1.3$$

Our fitted line is:

$$\boxed{\hat{y} = 1.3 + 0.9x}$$

Up to this point, we haven't used a normal distribution at all.

We have simply solved an optimization problem.

Even if your data are not normally distributed, you can calculate this line.

So why do we need probability?

2. A fitted slope is not necessarily the true slope

Imagine we collect data from five *different* students.

We will probably obtain different observations.

As a result, our fitted line might be:

$$\hat{y} = 1.8 + 0.7x$$

Or perhaps:

$$\hat{y} = 0.9 + 1.1x$$

See what's happening?

Even if there is one real relationship between studying and performance, different random samples produce different fitted slopes.

Therefore, we must distinguish two quantities.

Symbol	Meaning
β_1	True, unknown population slope
$\hat{\beta}_1$	Slope estimated from our sample

For example, perhaps:

$$\beta_1 = 1$$

But our five observations produce:

$$\hat{\beta}_1 = 0.9$$

Another sample could produce 1.2.

We don't know β_1, so we want to determine **how far our estimate might be from the true slope.**

And that's where probability enters.

3. Where does the normal distribution come from?

Here is the central modeling assumption of classical normal linear regression:

$$\boxed{Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i}$$

This equation has three parts:

- $\beta_0 + \beta_1 x_i$: the systematic relationship.
- ε_i: unpredictable variation.
- Y_i: the actual observed outcome.

For example, perhaps the real relationship is:

$$Y = 1 + X + \varepsilon$$

A student who studies 3 hours has a systematic predicted score of 4.

But perhaps the student slept badly and gets 2. Or perhaps the questions were familiar and the student gets 6.

We represent these unpredictable effects by ε.

The important normality assumption

We often assume:

$$\boxed{\varepsilon_i \sim N(0, \sigma^2)}$$

with independent errors sharing the same variance.

What does this mean intuitively?

At each fixed number of study hours, the scores fluctuate around the regression line.

Suppose the true expected score at $x = 3$ is 4.

Then, if $\sigma = 1$:

$$Y \mid X = 3 \sim N(4, 1)$$

At $x = 5$:

$$Y \mid X = 5 \sim N(6, 1)$$

Look at the following illustration.

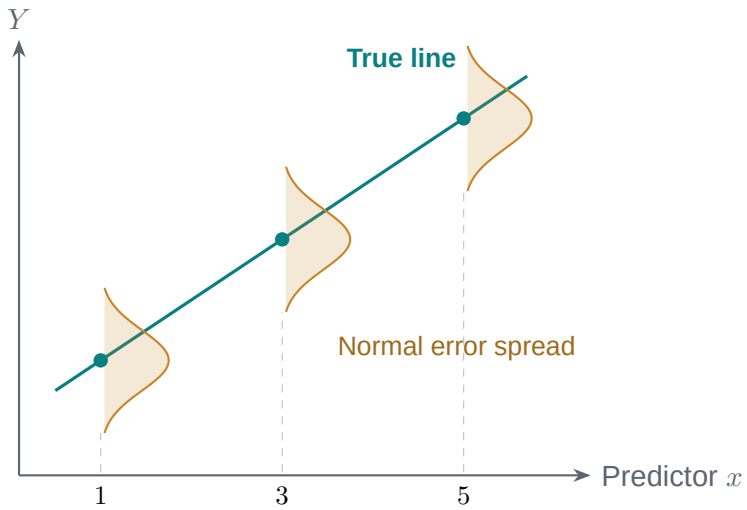


Figure 2: Conceptual illustration: at each fixed x, outcomes have a normal distribution centered on the true regression line. The orange sideways curves represent vertical spreads, not additional regression lines.

This is the connection you've been missing:

We are not assuming the entire scatterplot follows one normal distribution. We are assuming the errors around the line are normally distributed.

Equivalently:

$$\boxed{Y_i \mid x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)}$$

Notice that the mean of this normal distribution depends on x_i.

So we have many conditional normal distributions, each centered at a different height on the regression line.

4. Now the surprising part: the estimated slope is also normally distributed

Remember how we calculated the slope:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(Y_i - \bar{Y})}{\sum(x_i - \bar{x})^2}$$

This looks complicated, but we can rewrite it.

Define:

$$S_{xx} = \sum(x_i - \bar{x})^2$$

Then:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})Y_i}{S_{xx}}$$

Now substitute our regression model:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

We get:

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(\beta_0 + \beta_1 x_i + \varepsilon_i)}{S_{xx}}$$

Distribute the sum.

Because:

$$\sum(x_i - \bar{x}) = 0$$

the intercept term disappears.

And because:

$$\sum(x_i - \bar{x})x_i = S_{xx}$$

the slope term simplifies.

We obtain:

$$\boxed{\hat{\beta}_1 = \beta_1 + \frac{\sum(x_i - \bar{x})\varepsilon_i}{S_{xx}}}$$

This is the most important equation in this explanation.

Read it as:

$$\underbrace{\hat{\beta}_1}_{\text{Estimated slope}} = \underbrace{\beta_1}_{\text{True slope}} + \underbrace{\text{Random error}}_{\text{From our sample}}$$

Our estimated slope is the real slope plus a random disturbance.

Why does that make the slope normally distributed?

Because we assumed the errors ε_i are independent normal random variables.

And you already know this property of normal random variables:

A linear combination of independent normal random variables is normally distributed.

Our slope is precisely a linear combination of those errors, plus a constant.

Therefore:

$$\hat{\beta}_1 \sim N(\text{some mean}, \text{some variance})$$

Let's derive both.

5. Expected value and variance of the estimated slope

From our previous equation:

$$\hat{\beta}_1 = \beta_1 + \frac{\sum(x_i - \bar{x})\varepsilon_i}{S_{xx}}$$

Because the errors have mean zero:

$$E[\varepsilon_i] = 0$$

We get:

$$\boxed{E[\hat{\beta}_1] = \beta_1}$$

This means the least-squares slope estimator is **unbiased** under our assumptions.

Across repeated samples, its average equals the true slope.

Now calculate its variance.

Because the errors are independent and each has variance σ^2:

$$\text{Var}(\hat{\beta}_1) = \frac{\sum(x_i - \bar{x})^2\sigma^2}{S_{xx}^2}$$

Since the numerator contains $\sigma^2 S_{xx}$:

$$\boxed{\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}}$$

Therefore:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

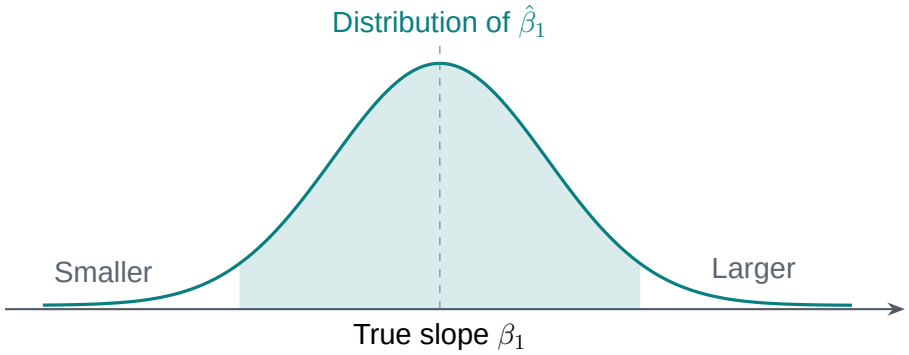


Figure 3: The distribution of estimated slopes across hypothetical repeated samples, keeping the same x-values and error model.

We're now very close to constructing a confidence interval.

But notice another interesting property.

The standard deviation of the estimated slope is:

$$\text{SD}(\hat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}}$$

If the observations have very different x-values, S_{xx} is large.

That makes the slope more precise.

Intuitively, two observations very close together provide little information about how steep a line is. Observations widely separated in x make its slope easier to estimate.

6. Finally, constructing a confidence interval for the slope

Recall the confidence interval for an ordinary population mean when σ is known:

$$\bar{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

Why did that work?

Because $\bar{X}$ had a known normal sampling distribution.

Exactly the same reasoning applies to the estimated regression slope.

We now have:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

Standardize:

$$Z = \frac{\hat{\beta}_1 - \beta_1}{\sigma/\sqrt{S_{xx}}}$$

Therefore:

$$Z \sim N(0,1)$$

We know:

$$P(-1.96 \leq Z \leq 1.96) = 0.95$$

Substitute the expression for Z and rearrange the inequalities to isolate β_1.

We get:

$$\boxed{CI_{\beta_1} = \hat{\beta}_1 \pm 1.96\frac{\sigma}{\sqrt{S_{xx}}}}$$

This is an exact 95% Z confidence interval for the slope when the normal-error model holds and the error standard deviation σ is known.

Notice the similarities:

Confidence interval for mean	**Confidence interval for slope**
Estimate: $\bar{X}$	Estimate: $\hat{\beta}_1$
True parameter: μ	True parameter: β_1
Standard error: $\sigma/\sqrt{n}$	Standard error: $\sigma/\sqrt{S_{xx}}$
Distribution: normal	Distribution: normal

It's the same statistical argument, applied to a different estimator.

7. Why Student's t-distribution appears

In reality, we almost never know σ, the true standard deviation of the regression errors.

So we estimate it using the residuals.

Recall that a residual is:

$$e_i = y_i - \hat{y}_i$$

We calculate:

$$RSS = \sum e_i^2$$

Then estimate error variance:

$$\boxed{s^2 = \frac{RSS}{n-2}}$$

Why $n-2$?

Because we estimated two unknown parameters:

- The intercept β_0.
- The slope β_1.

This leaves $n-2$ residual degrees of freedom.

Now the estimated standard error of the slope is:

$$\widehat{SE}(\hat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}$$

Instead of a standard normal distribution, the standardized statistic follows Student's t-distribution under the normal-error assumptions:

$$\boxed{T = \frac{\hat{\beta}_1 - \beta_1}{s/\sqrt{S_{xx}}} \sim t_{n-2}}$$

Hence:

$$\boxed{CI_{\beta_1} = \hat{\beta}_1 \pm t_{0.975,n-2}\frac{s}{\sqrt{S_{xx}}}}$$

This is the standard 95% confidence interval for a regression slope.

8. Complete example using our five students

We already calculated:

$$\hat{\beta}_1 = 0.9, \qquad \hat{\beta}_0 = 1.3$$

Our fitted line was:

$$\hat{y} = 1.3 + 0.9x$$

Let's calculate the residuals.

x	Observed y	Predicted $\hat{y}$	Residual
1	2	2.2	-0.2
2	3	3.1	-0.1
3	5	4.0	1.0
4	4	4.9	-0.9
5	6	5.8	0.2

Our sum of squared residuals is:

$$RSS = 0.04 + 0.01 + 1 + 0.81 + 0.04 = 1.9$$

With $n = 5$:

$$s^2 = \frac{1.9}{5-2} = \frac{1.9}{3}$$

$$s \approx 0.796$$

We already calculated:

$$S_{xx} = 10$$

Therefore:

$$\widehat{SE}(\hat{\beta}_1) = \frac{0.796}{\sqrt{10}} \approx 0.252$$

Our t-distribution has:

$$df = n - 2 = 3$$

For a 95% interval:

$$t_{0.975,3} \approx 3.182$$

The margin of error is:

$$ME = 3.182(0.252) \approx 0.801$$

Finally:

$$CI = 0.9 \pm 0.801$$

$$\boxed{CI_{\beta_1} \approx [0.099, 1.701]}$$

What does this interval tell us?

Our best estimate of the true slope is 0.9.

But our sample contains only five observations, and the residuals show some variation.

So there's significant uncertainty surrounding our estimate.

Under the normal linear regression assumptions, this confidence-interval procedure captures the true population slope in 95% of repeated samples.

The interval is quite wide because we have very little data.

Importantly, this interval concerns the **slope of the population relationship**, not an individual student's future score.

9. How the three concepts fit together

Step 1 — Linear regression
Find a line that minimizes squared prediction errors.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

An optimization problem.

⇓

Step 2 — Probability model
Assume independent normal errors around a true line.

$$\varepsilon_i \sim N(0, \sigma^2)$$

A probabilistic modeling assumption.

⇓

Step 3 — Sampling distribution
Derive the probability distribution of the fitted slope.

$$\hat{\beta}_1 \sim N(\beta_1, \sigma^2 / S_{xx})$$

A mathematical consequence of the model.

⇓

Step 4 — Statistical inference
Use the sampling distribution to quantify uncertainty.

$$\hat{\beta}_1 \pm t^* \widehat{SE}(\hat{\beta}_1)$$

A confidence interval for the true slope.

One subtle but important point: **normality is not required to calculate a least-squares line.** It is used here to derive an *exact finite-sample* t confidence interval. Other statistical methods can construct regression confidence intervals without normally distributed errors, under different assumptions.

10. A final question to check your intuition

Imagine that we've fitted a regression line to 100 students and found a slope of 2.

When we say the estimated slope is normally distributed, what exactly do we mean?

A. The 100 study-hour values follow a normal distribution.

B. The 100 exam scores follow a normal distribution.

C. If we repeated the experiment many times, the resulting estimated slopes would follow a normal distribution under the model assumptions.

D. The fitted regression line has a bell shape.

Answer: C.
The slope estimator is a random variable before we observe our data. Its distribution describes what happens to the slope across repeated samples. This is the distribution we use for confidence intervals.
The normality assumption concerns the errors around the true regression line, not necessarily the distribution of all x-values or all observed y-values. Those normal errors imply a normal sampling distribution for the fitted slope.

The deepest insight to retain is that there are two different random quantities here: the random errors that make observations fluctuate around the true line, and the random estimated slope that changes when we collect a new sample. The first creates the uncertainty; the second is what we use to measure that uncertainty.